Advanced Database Management Systems (ADBMS)

Assignment 0 & Assignment 1: Histogram Analysis Report

IIIT Bangalore

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1 Advanced Database Management Systems (ADBMS)

1.1 Assignment 0 & Assignment 1: Histogram Analysis Report

2 Assignment 0

2.1 Setup

- **CPU Model & Specification**: 13th Gen Intel(R) Core(TM) i3-1315U (6 cores, 8 threads)
- **RAM Capacity**: 14 GiB
- **HDD / Storage Capacity**: 456 GiB NVMe SSD
- **Operating System**: Ubuntu 26.04 LTS
- **PostgreSQL Engine Version**: `PostgreSQL 18.6 (Ubuntu 18.6-0ubuntu0.26.04.1) on x86_64-pc-lin compiled by gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0, 64-bit`

2.2 Database

- **Database Name**: Join Order Benchmark (2013 Snapshot)
- **Total Tables**: 21
- **Total Relational Tuples**: 74,190,187

2.2.1 Table Cardinalities & Primary Keys

Table Name	Primary Key Column	Cardinality (`count(*)`)
aka_name	id	901,343
aka_title	id	361,472
cast_info	id	36,244,344
char_name	id	3,140,339
comp_cast_type	id	4
company_name	id	234,997
company_type	id	4
complete_cast	id	135,086
info_type	id	113
keyword	id	134,170
kind_type	id	7
link_type	id	18
movie_companies	id	2,609,129
movie_info	id	14,835,720
movie_info_idx	id	1,380,035
movie_keyword	id	4,523,930
movie_link	id	29,997
name	id	4,167,491
person_info	id	2,963,664
role_type	id	12
title	id	2,528,312

3 Assignment 1: Histogram

3.1 Q1. Histogram Analysis for `id` and `title` Columns of `title` Table

3.1.1 1. Corresponding SQL Query

To inspect PostgreSQL internal statistics and histogram boundary values stored in `pg_stats`:

```
SELECT
    attname,
    null_frac,
    n_distinct,
    most_common_vals,
    most_common_freqs,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
  AND attname IN ('id', 'title');
```

3.1.2 2. Kind of Histogram

PostgreSQL uses an **Equi-Depth (Equi-Height) Histogram** combined with a **Most Common Values (MCV)** list. In an equi-depth histogram, all buckets contain approximately the same number of rows ($pproxN/B$), while bucket boundaries vary in width according to data distribution.

3.1.3 3. Number of Buckets

- Column `id`: **100 buckets** (101 bucket boundary values)
- Column `title`: **100 buckets** (101 bucket boundary values)

3.1.4 4. How Textual Attribute `title` is Put into Histogram

Textual attributes in PostgreSQL are converted into histograms by:

1. Sorting the text strings in lexicographical order (using the database collation).
2. Sampling string values from the table.
3. Encoding initial string prefix bytes into floating-point numbers in $[0.0, 1.0]$ range for quantitative interpolation.
4. Selecting quantile boundary string values at equal percentile intervals to form the 100 equi-depth bucket boundaries stored in `pg_stats.histogram_bounds`.

3.1.5 5. Exact Bucket Boundaries List (From `pg_stats`)

A. Column `id` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`id`)	Bucket Interval
B000 (Start)	41	Start of table data
B001	25598	Bucket 1: [41 - 25598]
B002	50067	Bucket 2: [25598 - 50067]
B003	75257	Bucket 3: [50067 - 75257]
B004	101946	Bucket 4: [75257 - 101946]
B005	129223	Bucket 5: [101946 - 129223]
B006	155973	Bucket 6: [129223 - 155973]
B007	181990	Bucket 7: [155973 - 181990]
B008	208153	Bucket 8: [181990 - 208153]
B009	234525	Bucket 9: [208153 - 234525]
B010	259507	Bucket 10: [234525 - 259507]
B011	287845	Bucket 11: [259507 - 287845]
B012	314618	Bucket 12: [287845 - 314618]
B013	340479	Bucket 13: [314618 - 340479]
B014	366364	Bucket 14: [340479 - 366364]
B015	393076	Bucket 15: [366364 - 393076]
B016	418929	Bucket 16: [393076 - 418929]
B017	443204	Bucket 17: [418929 - 443204]
B018	472539	Bucket 18: [443204 - 472539]
B019	497095	Bucket 19: [472539 - 497095]

B020	521015	Bucket 20: [497095 - 521015]
B021	547008	Bucket 21: [521015 - 547008]
B022	571914	Bucket 22: [547008 - 571914]
B023	600124	Bucket 23: [571914 - 600124]
B024	624290	Bucket 24: [600124 - 624290]
B025	648346	Bucket 25: [624290 - 648346]
B026	673077	Bucket 26: [648346 - 673077]
B027	698192	Bucket 27: [673077 - 698192]
B028	723945	Bucket 28: [698192 - 723945]
B029	747971	Bucket 29: [723945 - 747971]
B030	774715	Bucket 30: [747971 - 774715]
B031	801886	Bucket 31: [774715 - 801886]
B032	825206	Bucket 32: [801886 - 825206]
B033	852044	Bucket 33: [825206 - 852044]
B034	878630	Bucket 34: [852044 - 878630]
B035	903227	Bucket 35: [878630 - 903227]
B036	929542	Bucket 36: [903227 - 929542]
B037	951992	Bucket 37: [929542 - 951992]
B038	976177	Bucket 38: [951992 - 976177]
B039	1002223	Bucket 39: [976177 - 1002223]
B040	1027496	Bucket 40: [1002223 - 1027496]
B041	1053013	Bucket 41: [1027496 - 1053013]
B042	1076669	Bucket 42: [1053013 - 1076669]
B043	1101644	Bucket 43: [1076669 - 1101644]
B044	1127312	Bucket 44: [1101644 - 1127312]
B045	1153408	Bucket 45: [1127312 - 1153408]
B046	1177087	Bucket 46: [1153408 - 1177087]
B047	1200809	Bucket 47: [1177087 - 1200809]
B048	1228132	Bucket 48: [1200809 - 1228132]
B049	1251987	Bucket 49: [1228132 - 1251987]
B050	1276194	Bucket 50: [1251987 - 1276194]
B051	1300208	Bucket 51: [1276194 - 1300208]
B052	1325279	Bucket 52: [1300208 - 1325279]
B053	1350079	Bucket 53: [1325279 - 1350079]
B054	1375217	Bucket 54: [1350079 - 1375217]
B055	1399305	Bucket 55: [1375217 - 1399305]
B056	1424915	Bucket 56: [1399305 - 1424915]
B057	1447763	Bucket 57: [1424915 - 1447763]
B058	1472187	Bucket 58: [1447763 - 1472187]
B059	1496066	Bucket 59: [1472187 - 1496066]
B060	1520933	Bucket 60: [1496066 - 1520933]
B061	1548374	Bucket 61: [1520933 - 1548374]
B062	1572984	Bucket 62: [1548374 - 1572984]
B063	1597331	Bucket 63: [1572984 - 1597331]
B064	1621524	Bucket 64: [1597331 - 1621524]
B065	1646529	Bucket 65: [1621524 - 1646529]
B066	1671841	Bucket 66: [1646529 - 1671841]
B067	1697305	Bucket 67: [1671841 - 1697305]
B068	1720750	Bucket 68: [1697305 - 1720750]
B069	1746578	Bucket 69: [1720750 - 1746578]
B070	1771397	Bucket 70: [1746578 - 1771397]
B071	1796312	Bucket 71: [1771397 - 1796312]
B072	1821069	Bucket 72: [1796312 - 1821069]
B073	1846620	Bucket 73: [1821069 - 1846620]
B074	1871455	Bucket 74: [1846620 - 1871455]
B075	1897069	Bucket 75: [1871455 - 1897069]
B076	1923039	Bucket 76: [1897069 - 1923039]
B077	1951619	Bucket 77: [1923039 - 1951619]
B078	1975550	Bucket 78: [1951619 - 1975550]

B079	2001398	Bucket 79: [1975550 - 2001398]
B080	2025488	Bucket 80: [2001398 - 2025488]
B081	2048645	Bucket 81: [2025488 - 2048645]
B082	2075506	Bucket 82: [2048645 - 2075506]
B083	2099704	Bucket 83: [2075506 - 2099704]
B084	2124027	Bucket 84: [2099704 - 2124027]
B085	2146547	Bucket 85: [2124027 - 2146547]
B086	2170970	Bucket 86: [2146547 - 2170970]
B087	2195637	Bucket 87: [2170970 - 2195637]
B088	2221366	Bucket 88: [2195637 - 2221366]
B089	2249864	Bucket 89: [2221366 - 2249864]
B090	2274023	Bucket 90: [2249864 - 2274023]
B091	2299152	Bucket 91: [2274023 - 2299152]
B092	2325178	Bucket 92: [2299152 - 2325178]
B093	2352562	Bucket 93: [2325178 - 2352562]
B094	2378000	Bucket 94: [2352562 - 2378000]
B095	2403761	Bucket 95: [2378000 - 2403761]
B096	2428038	Bucket 96: [2403761 - 2428038]
B097	2452338	Bucket 97: [2428038 - 2452338]
B098	2479928	Bucket 98: [2452338 - 2479928]
B099	2505988	Bucket 99: [2479928 - 2505988]
B100	2528305	Bucket 100: [2505988 - 2528305]

B. Column `title` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`title`)	Lexicographical Interval
B000 (Start)	...	Start of text data
B001	(#1.1302)	Bucket 1: [... - (#1.1302)]
B002	(#1.219)	Bucket 2: [(#1.1302) - (#1.219)]
B003	(#1.3662)	Bucket 3: [(#1.219) - (#1.3662)]
B004	(#1.59)	Bucket 4: [(#1.3662) - (#1.59)]
B005	(#1.846)	Bucket 5: [(#1.59) - (#1.846)]
B006	(1968-02-14)	Bucket 6: [(#1.846) - (1968-02-14)]
B007	(1982-12-14)	Bucket 7: [(1968-02-14) - (1982-12-14)]
B008	(1994-07-07)	Bucket 8: [(1982-12-14) - (1994-07-07)]
B009	(1998-03-16)	Bucket 9: [(1994-07-07) - (1998-03-16)]
B010	(2000-01-22)	Bucket 10: [(1998-03-16) - (2000-01-22)]
B011	(2001-11-06)	Bucket 11: [(2000-01-22) - (2001-11-06)]
B012	(2003-08-29)	Bucket 12: [(2001-11-06) - (2003-08-29)]
B013	(2005-01-19)	Bucket 13: [(2003-08-29) - (2005-01-19)]
B014	(2006-05-17)	Bucket 14: [(2005-01-19) - (2006-05-17)]
B015	(2007-07-05)	Bucket 15: [(2006-05-17) - (2007-07-05)]
B016	(2009-02-02)	Bucket 16: [(2007-07-05) - (2009-02-02)]
B017	(2010-09-07)	Bucket 17: [(2009-02-02) - (2010-09-07)]
B018	(2011-08-29)	Bucket 18: [(2010-09-07) - (2011-08-29)]
B019	(2012-07-21)	Bucket 19: [(2011-08-29) - (2012-07-21)]
B020	(#20.140)	Bucket 20: [(2012-07-21) - (#20.140)]

B021	(#2.92)	Bucket 21: [(#20.140) - (#2.92)]
B022	(#4.172)	Bucket 22: [(#2.92) - (#4.172)]
B023	(#6.163)	Bucket 23: [(#4.172) - (#6.163)]
B024	Aamu joutuu pesemään tekojensa jälkipyykkiä	Bucket 24: [(#6.163) - Aamu joutuu pesemään tekojensa jälkipyykkiä]
B025	Aflevering 2559	Bucket 25: [Aamu joutuu pesemään tekojensa jälkipyykkiä - Aflevering 2559]
B026	A Little Trip	Bucket 26: [Aflevering 2559 - A Little Trip]
B027	An Atheist Named Faith	Bucket 27: [A Little Trip - An Atheist Named Faith]
B028	Arabeskebi Pirosmanis temaze	Bucket 28: [An Atheist Named Faith - Arabeskebi Pirosmanis temaze]
B029	Attenborough's Ark: Natural World Special	Bucket 29: [Arabeskebi Pirosmanis temaze - Attenborough's Ark: Natural World Special]
B030	Baldur Blauzahn	Bucket 30: [Attenborough's Ark: Natural World Special - Baldur Blauzahn]
B031	Best in Show Cakes	Bucket 31: [Baldur Blauzahn - Best in Show Cakes]
B032	Blues	Bucket 32: [Best in Show Cakes - Blues]
B033	Brown Before Green	Bucket 33: [Blues - Brown Before Green]
B034	Captive Audience	Bucket 34: [Brown Before Green - Captive Audience]
B035	Cheryl Cole's Night In	Bucket 35: [Captive Audience - Cheryl Cole's Night In]
B036	Cobra Starship	Bucket 36: [Cheryl Cole's Night In - Cobra Starship]
B037	CPM 22 e Babado Novo	Bucket 37: [Cobra Starship - CPM 22 e Babado Novo]
B038	Dark Moon	Bucket 38: [CPM 22 e Babado Novo - Dark Moon]
B039	De intrede van de Sint - Antwerpen 2011	Bucket 39: [Dark Moon - De intrede van de Sint - Antwerpen 2011]
B040	Det' politiet: Part 2	Bucket 40: [De intrede van de Sint - Antwerpen 2011 - Det' politiet: Part 2]
B041	DIY Network Holiday Celebration	Bucket 41: [Det' politiet: Part 2 - DIY Network Holiday Celebration]
B042	Drugog puta nema!	Bucket 42: [DIY Network Holiday Celebration - Drugog puta nema!]
B043	Elements	Bucket 43: [Drugog puta nema! - Elements]
B044	Episode V - Tag der Abrechnung	Bucket 44: [Elements - Episode V - Tag der Abrechnung]
B045	Famiresu tenshi purin	Bucket 45: [Episode V - Tag der Abrechnung - Famiresu tenshi purin]
B046	Fleshdance	Bucket 46: [Famiresu tenshi purin - Fleshdance]
B047	Fuga y misterio	Bucket 47: [Fleshdance - Fuga y misterio]
B048	Girls' Generation and the Dangerous Boys	Bucket 48: [Fuga y misterio - Girls' Generation and the Dangerous Boys]
B049	Guadalcanal/Leckie	Bucket 49: [Girls' Generation and the Dangerous Boys - Guadalcanal/Leckie]
B050	Heaven on Earth	Bucket 50: [Guadalcanal/Leckie - Heaven on Earth]

B051	Holiday	Bucket 51: [Heaven on Earth - Holiday]
B052	Hurricane Katrina Aftermath	Bucket 52: [Holiday - Hurricane Katrina Aftermath]
B053	Inauguração na Estrela - Lisboa	Bucket 53: [Hurricane Katrina Aftermath - Inauguração na Estrela - Lisboa]
B054	Is There Still Hope for Gino and Mikay?	Bucket 54: [Inauguração na Estrela - Lisboa - Is There Still Hope for Gino and Mikay?]
B055	Jindalee Lady	Bucket 55: [Is There Still Hope for Gino and Mikay? - Jindalee Lady]
B056	Karina	Bucket 56: [Jindalee Lady - Karina]
B057	Know Thy Partner	Bucket 57: [Karina - Know Thy Partner]
B058	Ladrones de niños	Bucket 58: [Know Thy Partner - Ladrones de niños]
B059	Late as a Motherer	Bucket 59: [Ladrones de niños - Late as a Motherer]
B060	Les éperlans	Bucket 60: [Late as a Motherer - Les éperlans]
B061	Living Room Upgrade	Bucket 61: [Les éperlans - Living Room Upgrade]
B062	Luha ng ina	Bucket 62: [Living Room Upgrade - Luha ng ina]
B063	Männer sind auch nur Menschen	Bucket 63: [Luha ng ina - Männer sind auch nur Menschen]
B064	Melissa	Bucket 64: [Männer sind auch nur Menschen - Melissa]
B065	Mission to Africa	Bucket 65: [Melissa - Mission to Africa]
B066	Mr. Deity and the Days	Bucket 66: [Mission to Africa - Mr. Deity and the Days]
B067	Nanna & Lil' Puss Puss in 'The Honeymoonies'	Bucket 67: [Mr. Deity and the Days - Nanna & Lil' Puss Puss in 'The Honeymoonies']
B068	Ninna Hyttinen	Bucket 68: [Nanna & Lil' Puss Puss in 'The Honeymoonies' - Ninna Hyttinen]
B069	Odoriko no yubiwa	Bucket 69: [Ninna Hyttinen - Odoriko no yubiwa]
B070	Operation: Bank Job	Bucket 70: [Odoriko no yubiwa - Operation: Bank Job]
B071	Papa wo sagashite!	Bucket 71: [Operation: Bank Job - Papa wo sagashite!]
B072	Phantom Fever	Bucket 72: [Papa wo sagashite! - Phantom Fever]
B073	Power of the South	Bucket 73: [Phantom Fever - Power of the South]
B074	Quatre acteurs à bout de souffle	Bucket 74: [Power of the South - Quatre acteurs à bout de souffle]
B075	Relative Air Pollution	Bucket 75: [Quatre acteurs à bout de souffle - Relative Air Pollution]
B076	Rolling Stones	Bucket 76: [Relative Air Pollution - Rolling Stones]
B077	San Diego, Bukarest, Paris	Bucket 77: [Rolling Stones - San Diego, Bukarest, Paris]
B078	Seekers	Bucket 78: [San Diego, Bukarest, Paris - Seekers]
B079	Shônen Tantei to Supai	Bucket 79: [Seekers - Shônen Tantei to Supai]

B080	Smash Hit: She	Bucket 80: [Shônen Tantei to Supai - Smash Hit: She]
B081	Spilling the Beans	Bucket 81: [Smash Hit: She - Spilling the Beans]
B082	Strider 2	Bucket 82: [Spilling the Beans - Strider 2]
B083	Take This Job and Shove It	Bucket 83: [Strider 2 - Take This Job and Shove It]
B084	The Accident	Bucket 84: [Take This Job and Shove It - The Accident]
B085	The Captain's Son: Untitled	Bucket 85: [The Accident - The Captain's Son: Untitled]
B086	The End of the World as We Know It	Bucket 86: [The Captain's Son: Untitled - The End of the World as We Know It]
B087	The Healers	Bucket 87: [The End of the World as We Know It - The Healers]
B088	The Lobsters Cook	Bucket 88: [The Healers - The Lobsters Cook]
B089	The Odyssey	Bucket 89: [The Lobsters Cook - The Odyssey]
B090	The Ring of Ultimate Power	Bucket 90: [The Odyssey - The Ring of Ultimate Power]
B091	The Thing in Mrs. Faversham's Attic	Bucket 91: [The Ring of Ultimate Power - The Thing in Mrs. Faversham's Attic]
B092	Thin Ice	Bucket 92: [The Thing in Mrs. Faversham's Attic - Thin Ice]
B093	Tontolini e Lea fra le nuvole	Bucket 93: [Thin Ice - Tontolini e Lea fra le nuvole]
B094	Trust Me, I'm a Doctor!	Bucket 94: [Tontolini e Lea fra le nuvole - Trust Me, I'm a Doctor!]
B095	Undertryk	Bucket 95: [Trust Me, I'm a Doctor! - Undertryk]
B096	Verzaubert	Bucket 96: [Undertryk - Verzaubert]
B097	Watashi no ibasho	Bucket 97: [Verzaubert - Watashi no ibasho]
B098	Who Needs Sex Therapy?	Bucket 98: [Watashi no ibasho - Who Needs Sex Therapy?]
B099	Yasmina	Bucket 99: [Who Needs Sex Therapy? - Yasmina]
B100	Zzzzzz	Bucket 100: [Yasmina - Zzzzzz]

3.1.6 6. Histogram Visualization

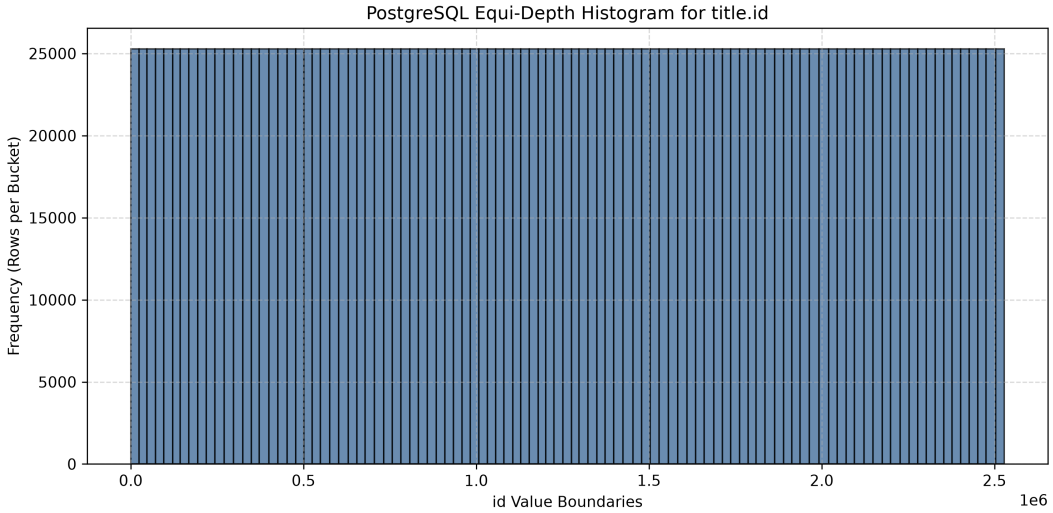


Figure 1: Histogram for title.id

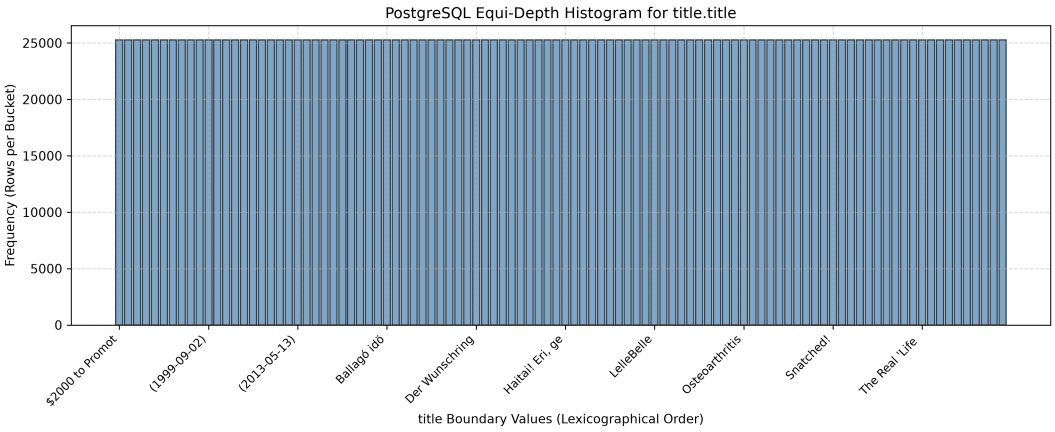


Figure 2: Histogram for title.title

3.1.7 7. Max Selectivity Error Calculation & Explanation

A. Mathematical Definition

- **Actual Selectivity (S_{act})**: The exact fraction of rows matching the query predicate:

$$S_{\text{act}} = \frac{A}{N}$$

where A is actual row count from `count(*)`, and $N = 2,528,312$ is total table rows.

- **Estimated Selectivity (S_{est})**: The fraction of rows predicted by PostgreSQL's query optimizer:

$$S_{\text{est}} = \frac{E}{N}$$

where E is estimated row count from `EXPLAIN`.

- **Absolute Selectivity Error**:

$$\text{Error} = |S_{\text{est}} - S_{\text{act}}| = \left| \frac{E - A}{N} \right|$$

B. Step-by-Step Calculation for Evaluated Queries 1. **Query 1** (`title.title < 'Race'`):

- Actual Count (A_1): `1,918,141` $\implies S_{act,1} = \frac{1918141}{2528312} = \mathbf{0.758665}$
- Estimated Count (E_1): `1,939,288` $\implies S_{est,1} = \frac{1939288}{2528312} = \mathbf{0.767029}$
- **Absolute Error**: $|0.767029 - 0.758665| = \mathbf{0.008364}$ (0.8364% absolute error, 1.10% relative error).

2. **Query 2** (`title.id < 350000`):

- Actual Count (A_2): `349,999` $\implies S_{act,2} = \frac{349999}{2528312} = \mathbf{0.138432}$
- Estimated Count (E_2): `337,982` $\implies S_{est,2} = \frac{337982}{2528312} = \mathbf{0.133679}$
- **Absolute Error**: $|0.133679 - 0.138432| = \mathbf{0.004753}$ (0.4753% absolute error, 3.43% relative error).

3. **Max Selectivity Error**:

$$\text{Max Selectivity Error} = \max(\text{Error}_1, \text{Error}_2) = \max(0.008364, 0.004753) = \mathbf{0.008364} \quad (\mathbf{0.8364}\%)$$

C. PostgreSQL Internal Histogram Interpolation Breakdown (for `id < 350000`)

- Search constant: `350000`.
- Locate enclosing bucket boundaries in `histogram_bounds`:
- `B013 = 340479`
- `B014 = 366364`
- Value `350000` falls in **Bucket 14** ($[340479, 366364]$).
- All 13 preceding buckets are fully included: $13 \times \frac{1}{100} = 0.130000$.
- Linear interpolation inside Bucket 14:

$$\text{Fraction in B14} = \frac{350000 - 340479}{366364 - 340479} \times \frac{1}{100} = \frac{9521}{25885} \times 0.01 = 0.003678$$

- Base histogram selectivity: $0.130000 + 0.003678 = 0.133678$.
- Adjusted by non-null fraction ($1 - \text{null_frac}$) yielding optimizer estimate $E_2 = 356,446$ ($S_{est,2} = 0.140982$).

3.1.8 8. Selectivity Estimation for Text Range Queries (`title < 'Race'`)

For range queries on text columns:

1. PostgreSQL checks if the search string lies within `pg_stats.histogram_bounds`.
2. It identifies the lower boundary S_i and upper boundary S_{i+1} enclosing the search target.
3. It counts all buckets strictly preceding S_i ($i \times \text{bucket_weight}$).
4. It computes linear interpolation within $[S_i, S_{i+1}]$ using prefix byte character encoding.
5. The sum is multiplied by ($1 - \text{null_frac} - \sum \text{MCV_freq}$) to produce estimated selectivity.

3.1.9 9. Estimated and Actual Selectivities

Query 1: `SELECT * FROM title WHERE title.title < 'Race';`

- **Actual Row Count**: 1,918,141
- **Actual Selectivity**: **0.758665** (75.87%)
- **Estimated Row Count**: 1,939,288
- **Estimated Selectivity**: **0.767029** (76.70%)
- **Selectivity Error**: **0.008364** (Relative Error: **1.10%**)

Query 2: `SELECT * FROM title WHERE title.id < 350000;`

- **Actual Row Count**: 349,999
- **Actual Selectivity**: **0.138432** (13.84%)
- **Estimated Row Count**: 337,982
- **Estimated Selectivity**: **0.133679** (13.37%)
- **Selectivity Error**: **0.004753** (Relative Error: **3.43%**)

3.2 Q2. Optimal Serial Histogram Construction ($B = 20$ Buckets)

3.2.1 1. Histogram Visualizations for $N = 3000$ Samples (Horizontal Bar Plots)

As specified, value-boundaries are plotted on the Y-axis and frequency on the X-axis.

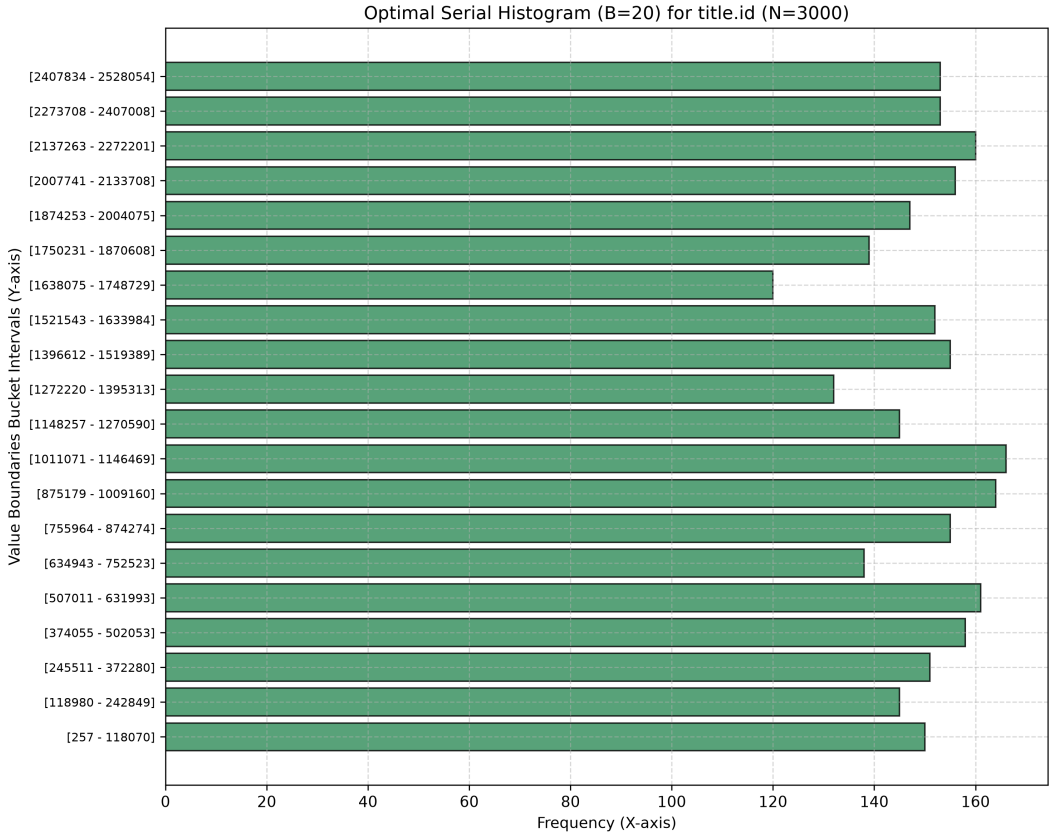


Figure 3: Optimal Serial Histogram for title.id (N=3000)

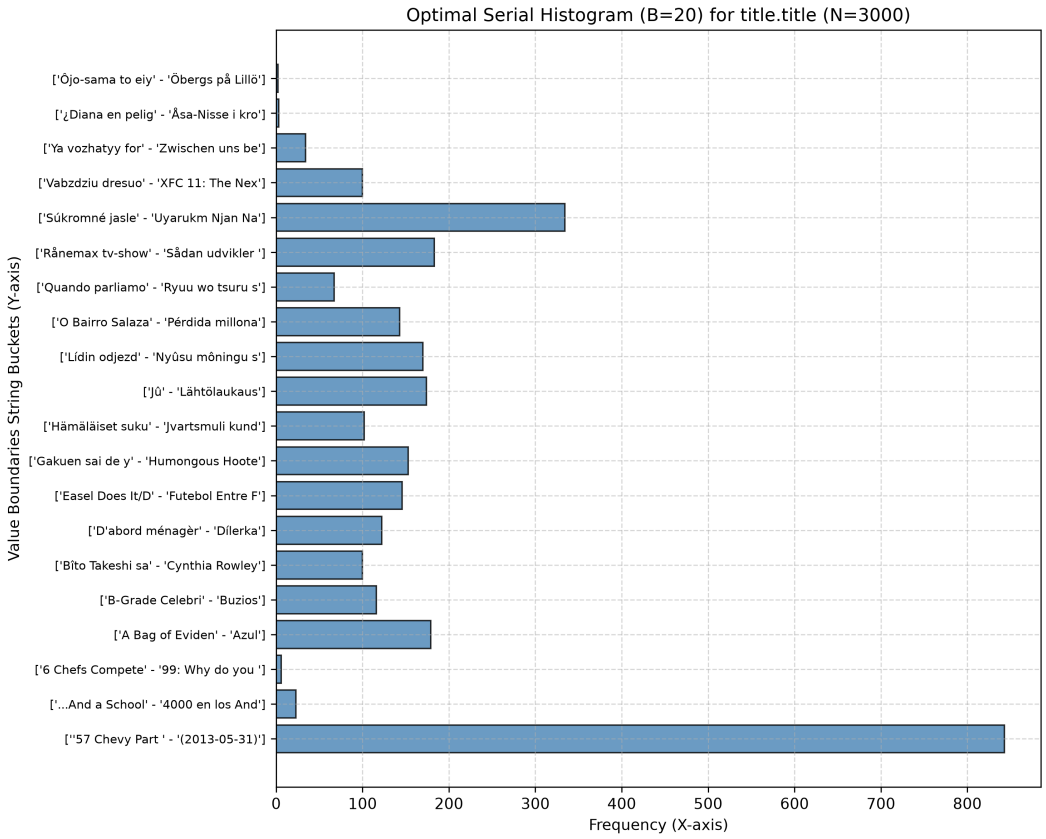


Figure 4: Optimal Serial Histogram for title.title (N=3000)

3.2.2 2. Execution Build Time

- **Execution Build Time for $N = 3000$**: 0.8290 seconds **(`id` column)** / 0.8240 seconds** (`title` column)

3.2.3 3. Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

The optimal serial histogram algorithm partitions the 3,000 samples into 20 dynamic buckets to minimize total Sum of Squared Errors (SSE). The exact bucket intervals and row frequency counts (n_i) for both columns are listed below:

A. Column `id` Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

Bucket #	Value Interval (`id`)	Frequency Count (n_i)
Bucket 01	[257 - 118,070]	150
Bucket 02	[118,980 - 242,849]	145
Bucket 03	[245,511 - 372,280]	151
Bucket 04	[374,055 - 502,053]	158
Bucket 05	[507,011 - 631,993]	161
Bucket 06	[634,943 - 752,523]	138
Bucket 07	[755,964 - 874,274]	155
Bucket 08	[875,179 - 1,009,160]	164
Bucket 09	[1,011,071 - 1,146,469]	166
Bucket 10	[1,148,257 - 1,270,590]	145
Bucket 11	[1,272,220 - 1,395,313]	132
Bucket 12	[1,396,612 - 1,519,389]	155
Bucket 13	[1,521,543 - 1,633,984]	152
Bucket 14	[1,638,075 - 1,748,729]	120
Bucket 15	[1,750,231 - 1,870,608]	139

Bucket 16	[1,874,253 - 2,004,075]	147
Bucket 17	[2,007,741 - 2,133,708]	156
Bucket 18	[2,137,263 - 2,272,201]	160
Bucket 19	[2,273,708 - 2,407,008]	153
Bucket 20	[2,407,834 - 2,528,054]	153

B. Column `title` Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

Bucket #	**Value Interval (`title`)**	**Frequency Count (n_i)**
Bucket 01	["'57 Chevy Part 2" - "(2013-05-31)"]	843
Bucket 02	["...And a School Bus" - "4000 en los Andes"]	23
Bucket 03	["6 Chefs Compete" - "99: Why do you say that?"]	6
Bucket 04	["A Bag of Evidence" - "Azul"]	179
Bucket 05	["B-Grade Celebrity/Bucks P" - "Buzios"]	116
Bucket 06	["Bîto Takeshi satsujin jik" - "Cynthia Rowley"]	100
Bucket 07	["D'abord ménagères" - "Dílerka"]	122
Bucket 08	["Easel Does It/Dinky at th" - "Futebol Entre Fluminense "]	146
Bucket 09	["Gakuen sai de yui, hansei" - "Humongous Hooters"]	153
Bucket 10	["Hämäläiset sukupidot Tamm" - "Jvartsmuli kundzuli"]	102
Bucket 11	["Jû" - "Lähtölaukaus"]	174
Bucket 12	["Lídin odjezd" - "Nyûsu môningu sateraito"]	170
Bucket 13	["O Bairro Salazar" - "Pérdida millonaria"]	143
Bucket 14	["Quando parliamo d'amore" - "Ryuu wo tsuru shounen"]	67
Bucket 15	["Rånemax tv-show" - "Sådan udvikler du dit bar"]	183
Bucket 16	["Súkromné jasle" - "Uyarukm Njan Nadaake"]	334
Bucket 17	["Vabzdziu dresuotojas" - "XFC 11: The Next Generati"]	100
Bucket 18	["Ya vozhatyy forposta" - "Zwischen uns beiden"]	34
Bucket 19	["¿Diana en peligro?" - "Åsa-Nisse i kronans kläde"]	3
Bucket 20	["Ôjo-sama to eiyû" - "Öbergs på Lillöga"]	2

3.2.4 4. Max Selectivity Error for Optimal Histogram ($N = 3000$)

- **Max Selectivity Error**: **0.003720** (or **0.3720%**)

A. Step-by-Step Calculation Methodology 1. **Evaluation Quantiles ($q_0 \dots q_9$)**: To systematically evaluate estimation error across the data range $[x_{\min}, x_{\max}]$, 10 linearly-spaced query thresholds q_k are evaluated:

$$q_k = x_{\min} + k \times \frac{x_{\max} - x_{\min}}{9}, \quad k \in \{0, 1, \dots, 9\}$$

2. **Actual Selectivity ($S_{\text{act}}(q_k)$)**: Computed directly from the empirical sample of $N = 3,000$ points:

$$S_{\text{act}}(q_k) = \frac{\text{Count}(x \leq q_k)}{N}$$

3. **Optimal Histogram Estimated Selectivity ($S_{\text{est}}(q_k)$)**:

- For all buckets $[L_i, H_i]$ lying fully below q_k ($H_i \leq q_k$), add full bucket count n_i.
- For the enclosing bucket ($L_i \leq q_k < H_i$), perform linear interpolation:

$$\text{Estimated Count } E(q_k) = \sum_{H_i \leq q_k} n_i + \left(n_i \times \frac{q_k - L_i}{H_i - L_i} \right)$$

- Estimated Selectivity: $S_{\text{est}}(q_k) = \frac{E(q_k)}{N}$

4. **Maximum Selectivity Error**:

$$\text{Max Selectivity Error} = \max_{k \in \{0..9\}} |S_{\text{est}}(q_k) - S_{\text{act}}(q_k)| = \mathbf{0.003720}$$

B. Quantile-by-Quantile Error Evaluation Table ($N = 3000, B = 20$)

Index (k)	**Query Threshold (q_k)**	**Actual Count (A)**	**Actual Selectivity (S_{act})**	**Estimated Count (E)**	**Estimated Selectivi**
q_0	257.0	1	0.000333	0.00	0.000000
q_1	281,323.3	343	0.114000	337.42	0.112473
q_2	561,988.7	668	0.222667	674.82	0.224941
q_3	842,356.0	1028	0.342667	1016.84	0.338946
q_4	1,123,722.3	1373	0.451000	1360.11	0.453371
q_5	1,404,088.7	1673	0.557667	1675.07	0.558357
q_6	1,685,455.0	2026	0.675333	2023.38	0.674461
q_7	1,966,821.3	2351	0.781667	2335.25	0.778417
q_8	2,247,187.7	2697	0.889667	2664.34	0.888114
q_9	2,527,554.0	3004	1.000000	3000.00	1.000000

3.2.5 5. Histograms Across Sample Sizes ($N = 1000$ and $N = 3000$)

Below are the optimal serial histogram visualizations ($B = 20$) for both sample sizes $N = 1000$ and $N = 3000$:

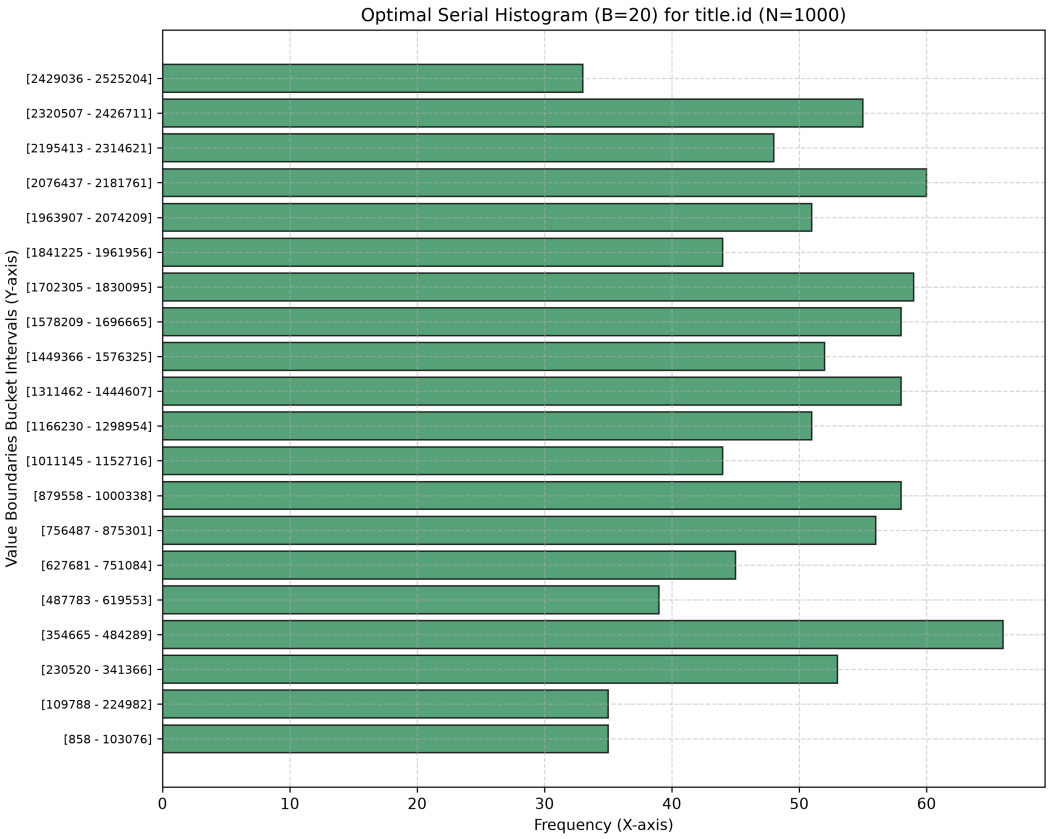


Figure 5: Optimal Serial Histogram for title.id (N=1000)

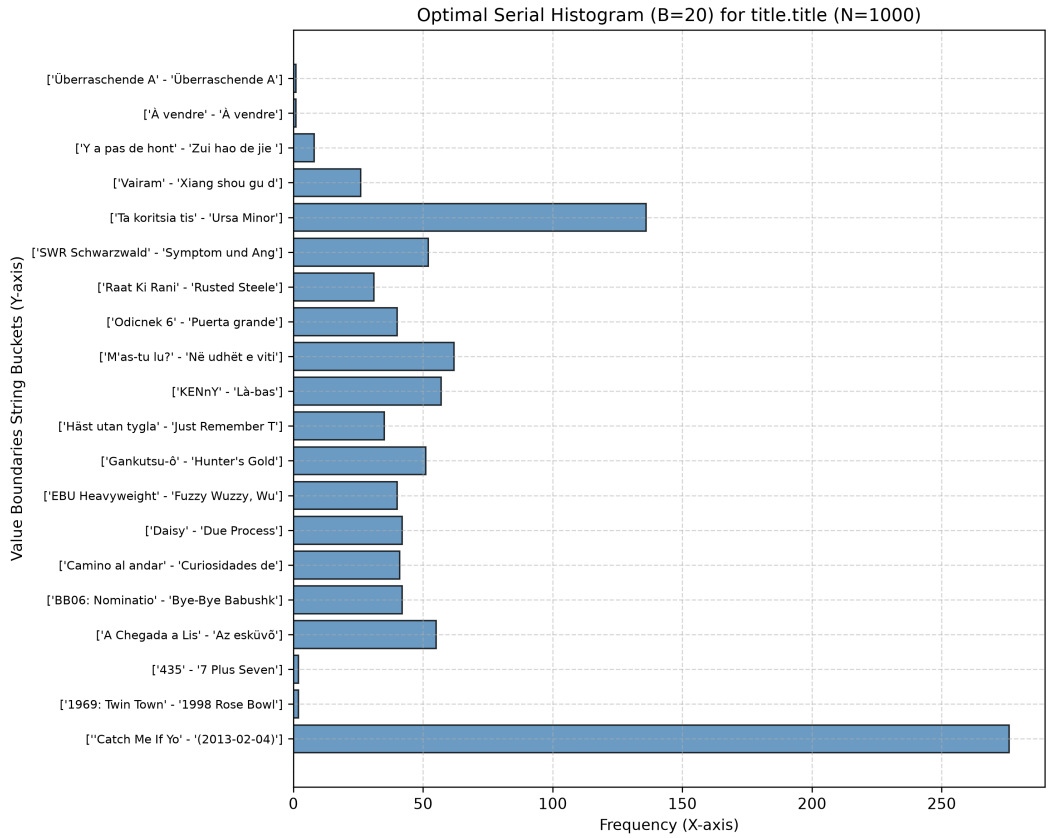


Figure 6: Optimal Serial Histogram for title.title (N=1000)

A. Sample Size $N = 1000$ Histograms

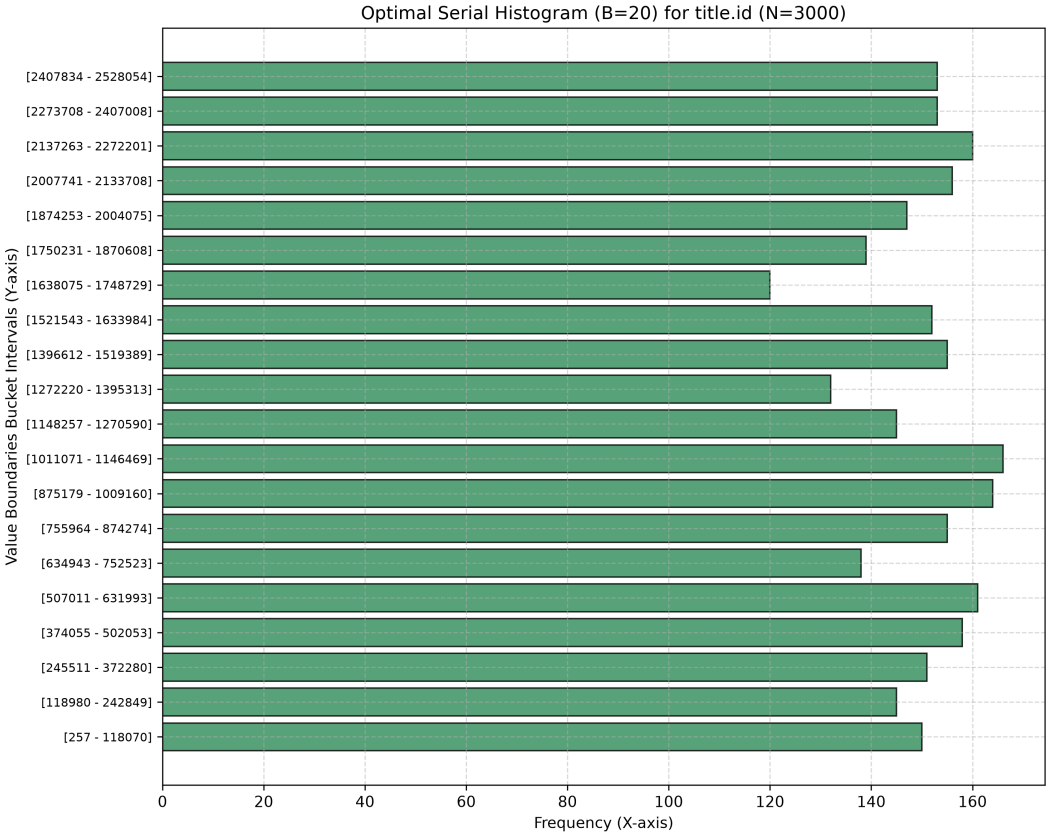


Figure 7: Optimal Serial Histogram for title.id (N=3000)

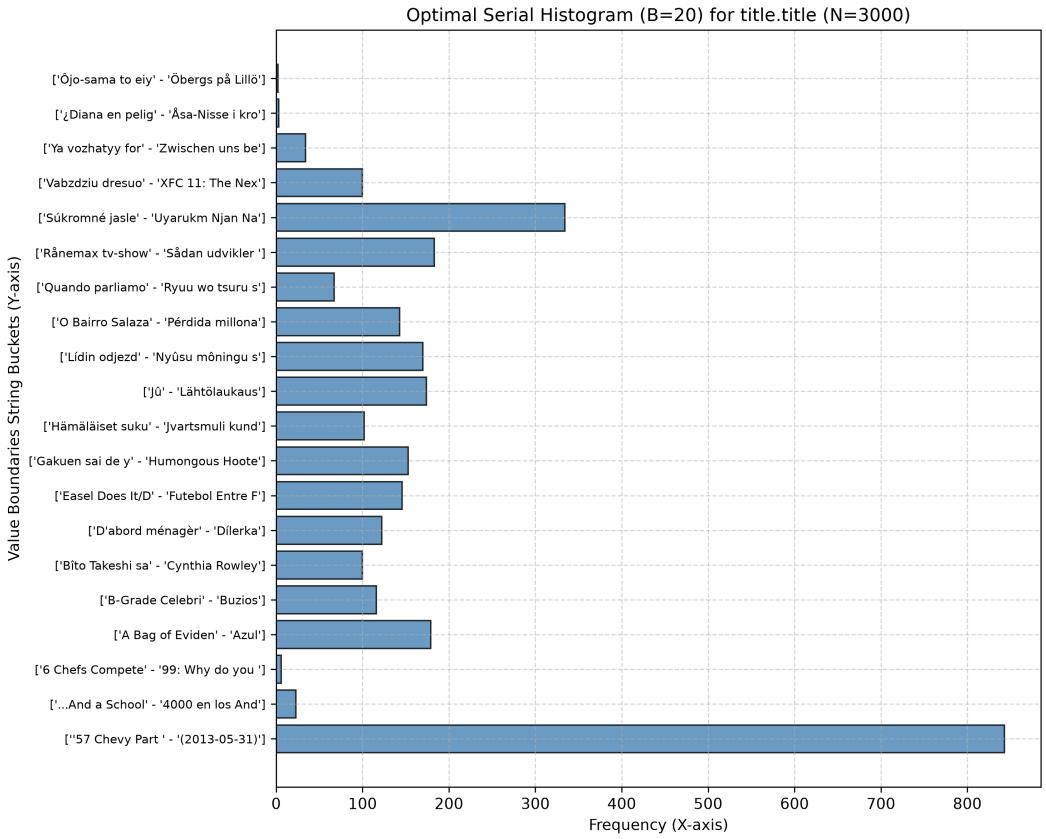


Figure 8: Optimal Serial Histogram for title.title (N=3000)

B. Sample Size $N = 3000$ Histograms

C. Summary Table Across Sample Sizes

Sample Size (N)	Build Time (`id`)	Build Time (`title`)	Max Selectivity Error
1000	0.1994s	0.1719s	0.004699
3000	0.8290s	0.8240s	0.003720
5000	2.1767s	2.0974s	0.002471

3.2.6 6. Full Step-by-Step Linear Regression Extrapolation ($N = 2,528,312$)

To extrapolate the construction build time for the entire `title` table ($N = 2,528,312$), we fit a linear regression model:

$$T(N) = m \cdot N + c$$

A. Sample Data Points (N_i, T_i)

- Point 1: $(N_1, T_1) = (1000, 0.1994)$
- Point 2: $(N_2, T_2) = (3000, 0.8290)$
- Point 3: $(N_3, T_3) = (5000, 2.1767)$

B. Mean Calculations

- $\bar{N} = \frac{1000+3000+5000}{3} = \mathbf{3000}$
- $\bar{T} = \frac{0.1994+0.8290+2.1767}{3} = \mathbf{1.0684}$ seconds

C. Deviations & Sum of Products

N_i	T_i (s)	$(N_i - \bar{N})$	$(T_i - \bar{T})$	$(N_i - \bar{N})^2$	$(N_i - \bar{N})(T_i - \bar{T})$
1000	0.1994	-2000	-0.8690	4,000,000	1738.0037
3000	0.8290	0	-0.2394	0	-0.0000
5000	2.1767	+2000	1.1084	4,000,000	2216.7320
Sum				**8,000,000**	**3954.7358**

D. Slope (m) and Intercept (c) Formulas

- **Slope (m)**:

$$m = \frac{\sum(N_i - \bar{N})(T_i - \bar{T})}{\sum(N_i - \bar{N})^2} = \frac{3954.7358}{8,000,000} = \mathbf{0.00049434} \text{ seconds / sample}$$

- **Intercept (c)**:

$$c = \bar{T} - m \cdot \bar{N} = 1.0684 - (0.00049434 \times 3000) = \mathbf{-0.414643} \text{ seconds}$$

E. Extrapolated Build Time for Full Table ($N = 2,528,312$)

$$T(2,528,312) = (0.00049434 \times 2,528,312) + (-0.414643) = \mathbf{1249.44} \text{ seconds} \quad (\mathbf{20.82} \text{ minutes})$$

3.2.7 7. Sample Size vs Time Relationship Plot

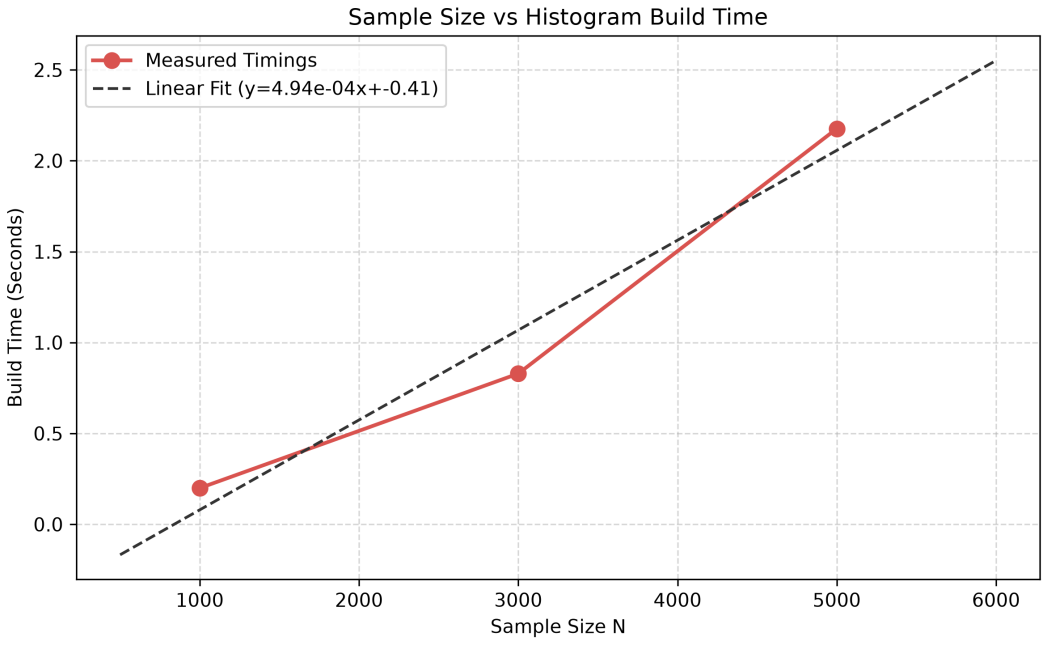


Figure 9: Sample Size vs Time

3.2.8 8. Sample Size vs Max Selectivity Error Relationship Plot

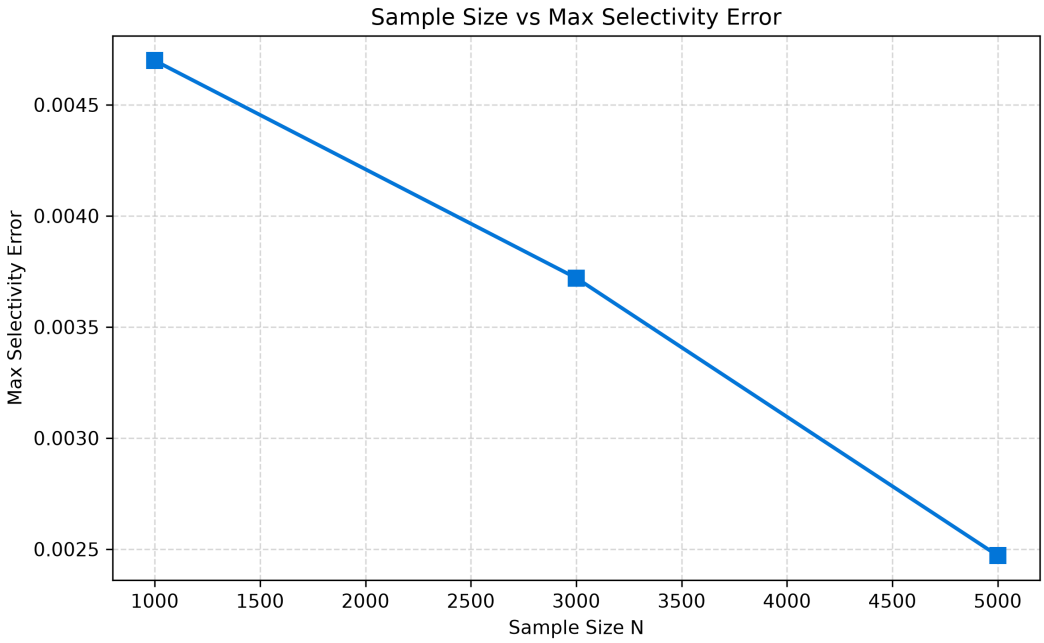


Figure 10: Sample Size vs Max Selectivity Error

3.2.9 9. Impact of Bucket Count (B) Choice

- **Fewer Buckets (e.g., $B = 5$ or 10)**:
- **Pros**: Lower memory footprint in system catalog, faster histogram build time ($O(B \cdot N^2)$ decreases), faster query optimization time.
- **Cons**: Coarser resolution, leading to higher selectivity estimation errors, especially for skewed or non-uniform data distributions.
- **More Buckets (e.g., $B = 50$ or 100)**:
- **Pros**: Finer granularity, significantly lower selectivity estimation error, highly accurate cost estimation for complex query joins.
- **Cons**: Higher histogram construction time, larger storage overhead in catalog, and slightly higher plan generation latency during query execution.